1. Calculate the power used by a refrigerator connected to 120 V with a current of 3.8 A.

Solution:

Given: V = 120 V I = 3.8 A

Unknown: P = ?

Equation: P = IV

Solution:

P = (120 V)(3.8 A) = 456 W

2. How much resistance is there in a 1200 W heater connected to a 120 V source?

Solution:

Given V = 120 V P = 1200 W

Unknown: R = ?

Equation: $P = \frac{V^2}{R}$

Solution:

$$R = \frac{V^2}{P}$$
$$R = \frac{(120V)^2}{1200W}$$
$$R = 12\Omega$$

3. A 24 W compact fluorescent light bulb has a 600 Ω resistance. Find the current in the light bulb.

Solution:

Given: R = 600 Ω P = 24 W

Unknown: I = ?

Equation: $P = I^2R$

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Solution:

$$I = \sqrt{\frac{P}{R}}$$
$$I = \sqrt{\frac{24W}{600\Omega}}$$
$$I = 0.2A$$

4.

Solution:

Step 1: This is a parallel circuit, so the electric potential everywhere is equal.

$$V_{battery} = V_1 = V_2$$

Step 2: Use Ohm's Law to find the current in R_2.

Step 3: Find the current through the battery. Look at the junction above R_2. You know 0.20 A flows down to R_2 and 0.10 A flows to R_1. Therefore 0.30 A must be flowing into the junction, because it splits into 0.10 A and 0.20 A.

	Current (A)	Electric potential (V)	Resistance (Ω)
Battery	**0.30 A**	**10 V**	
R_1	0.10 A	**10 V**	
R_2	**0.20 A**	10 V	50 Ω

b. Now use the current and electric potential at the battery to find the power in the circuit.

Solution:

Given: I = 0.30 A V = 10.0 V

Unknown: P = ?

Equation: $P = VI$

Solution:

P = (0.30 A)(10 V) = 3 W.

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